



MIT Art Design and Technology University
MIT School of Computing, Pune

**Department of Computer Science and
Engineering**

Assignment - 2

Subject - Data Visualization

Class - T.Y. AIEC-1

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Roll: 26**

A.Y. 2025 - 2026

1. Download a small CSV dataset (Kaggle / open data).

- Identify data quality issues
- Perform **data cleaning and preprocessing**
- Show **before and after** snapshots

R-CODE:

```

1 # Load dataset
2 df <- read.csv("C:\\Users\\Prithviraj\\Downloads\\Titanic-Dataset.csv")
3
4 # Before snapshot
5 head(df)
6
7 # Handle missing values
8 df$Age[is.na(df$Age)] <- median(df$Age, na.rm = TRUE)
9 df$Embarked[is.na(df$Embarked)] <- "S"
10
11 # Drop Cabin (too many missing)
12 df$Cabin <- NULL
13
14 # After snapshot
15 print(df)
16
17 write.csv(df, "C:/Users/Prithviraj/Downloads/cleaned_titanic.csv", row.names = FALSE)

```

Before Processing & Cleaning:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22.00	1	0	A/5 21171	7.2500	NA	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38.00	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.00	0	0	STON/O2. 3101282	7.9250	NA	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.00	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.00	0	0	373450	8.0500	NA	S
6	0	3	Moran, Mr. James	male	NA	0	0	330877	8.4583	NA	Q
7	0	1	McCarthy, Mr. Timothy J	male	54.00	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2.00	3	1	349909	21.0750	NA	S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.00	0	2	347742	11.1333	NA	S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.00	1	0	237736	30.0708	NA	C
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.00	1	1	PP 9549	16.7000	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58.00	0	0	113783	26.5500	C103	S
13	0	3	Saunderscock, Mr. William Henry	male	20.00	0	0	A/5. 2151	8.0500	NA	S
14	0	3	Andersson, Mr. Anders Johan	male	39.00	1	5	347082	31.2750	NA	S
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.00	0	0	350406	7.8542	NA	S
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.00	0	0	248706	16.0000	NA	S
17	0	3	Rice, Master. Eugene	male	2.00	4	1	382652	29.1250	NA	Q
18	1	2	Williams, Mr. Charles Eugene	male	NA	0	0	244373	13.0000	NA	S
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31.00	1	0	345763	18.0000	NA	S
20	1	3	Maselmami, Mrs. Fatima	female	NA	0	0	2649	7.2250	NA	C
21	0	2	Fynney, Mr. Joseph J	male	35.00	0	0	239865	26.0000	NA	S
22	1	2	Beesley, Mr. Lawrence	male	34.00	0	0	248698	13.0000	D56	S
23	1	3	McGowan, Miss. Anna "Annie"	female	15.00	0	0	330923	8.0292	NA	Q
24	1	1	Sloper, Mr. William Thompson	male	28.00	0	0	113788	35.5000	A6	S
25	0	3	Palsson, Miss. Torborg Danira	female	8.00	3	1	349909	21.0750	NA	S
26	1	3	Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johansson)	female	38.00	1	5	347077	31.3875	NA	S
27	0	3	Emir, Mr. Farred Chehab	male	NA	0	0	2631	7.2250	NA	C
28	0	1	Fortune, Mr. Charles Alexander	male	19.00	3	2	19950	263.0000	C23 C25 C27	S
29	1	3	O'Dwyer, Miss. Ellen "Nellie"	female	NA	0	0	330959	7.8792	NA	Q
30	0	3	Todoroff, Mr. Lalio	male	NA	0	0	349216	7.8958	NA	S
31	0	1	Uruchurtu, Don. Manuel E	male	40.00	0	0	PC 17601	27.7208	NA	C

After Processing & Cleaning:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22.00	1	0	A/5 21171	7.2500	S
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3	1	3	Heikkinen, Miss. Laina	female	26.00	0	0	STON/O2. 3101282	7.9250	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.00	1	0	113803	53.1000	S
5	0	3	Allen, Mr. William Henry	male	35.00	0	0	373450	8.0500	S
6	0	3	Moran, Mr. James	male	28.00	0	0	330877	8.4583	Q
7	0	1	McCarthy, Mr. Timothy J	male	54.00	0	0	17463	51.8625	S
8	0	3	Palsson, Master. Gosta Leonard	male	2.00	3	1	349909	21.0750	S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.00	0	2	347742	11.1333	S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.00	1	0	237736	30.0708	C
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.00	1	1	PP 9549	16.7000	S
12	1	1	Bonnell, Miss. Elizabeth	female	58.00	0	0	113783	26.5500	S
13	0	3	Saunderscock, Mr. William Henry	male	20.00	0	0	A/5. 2151	8.0500	S
14	0	3	Andersson, Mr. Anders Johan	male	39.00	1	5	347082	31.2750	S
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.00	0	0	350406	7.8542	S
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.00	0	0	248706	16.0000	S
17	0	3	Rice, Master. Eugene	male	2.00	4	1	382652	29.1250	Q
18	1	2	Williams, Mr. Charles Eugene	male	28.00	0	0	244373	13.0000	S
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31.00	1	0	345763	18.0000	S
20	1	3	Masselemani, Mrs. Fatima	female	28.00	0	0	2649	7.2250	C
21	0	2	Fynney, Mr. Joseph J	male	35.00	0	0	239865	26.0000	S
22	1	2	Beesley, Mr. Lawrence	male	34.00	0	0	248698	13.0000	S
23	1	3	McGowan, Miss. Anna "Annie"	female	15.00	0	0	330923	8.0292	Q
24	1	1	Sloper, Mr. William Thompson	male	28.00	0	0	113788	35.5000	S
25	0	3	Palsson, Miss. Torborg Danira	female	8.00	3	1	349909	21.0750	S
26	1	3	Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johansson)	female	38.00	1	5	347077	31.3875	S
27	0	3	Emir, Mr. Farred Chehab	male	28.00	0	0	2631	7.2250	C
28	0	1	Fortune, Mr. Charles Alexander	male	19.00	3	2	19950	263.0000	S
29	1	3	O'Dwyer, Miss. Ellen "Nellie"	female	28.00	0	0	330959	7.8792	Q
30	0	3	Todoroff, Mr. Lallo	male	28.00	0	0	349216	7.8958	S
31	0	1	Uruchurtu, Don. Manuel E	male	40.00	0	0	PC 17601	27.7208	C

2. Apply **basic EDA techniques** and summarize insights.

```
> summary(df)
 PassengerId      Survived      Pclass      Name      Sex
 Min.   : 1.0      Min.   :0.0000   Min.   :1.000   Length:891   Length:891
 1st Qu.:223.5      1st Qu.:0.0000   1st Qu.:2.000   Class :character   Class :character
 Median :446.0      Median :0.0000   Median :3.000   Mode  :character   Mode  :character
 Mean   :446.0      Mean   :0.3838   Mean   :2.309
 3rd Qu.:668.5      3rd Qu.:1.0000   3rd Qu.:3.000
 Max.   :891.0      Max.   :1.0000   Max.   :3.000

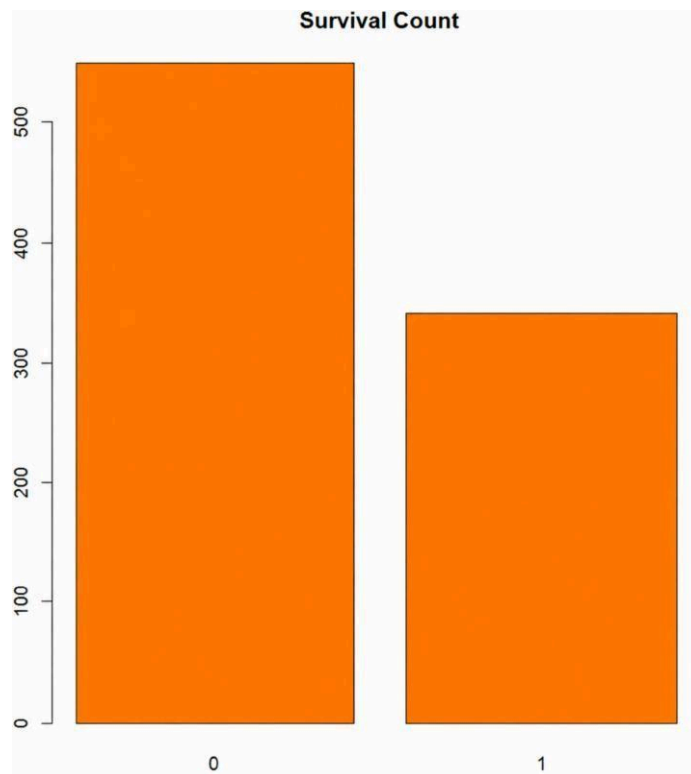
      Age      SibSp      Parch      Ticket      Fare
 Min.   : 0.42   Min.   :0.0000   Min.   :0.0000   Length:891   Min.   : 0.00
 1st Qu.:22.00   1st Qu.:0.0000   1st Qu.:0.0000   Class :character   1st Qu.: 7.91
 Median :28.00   Median :0.0000   Median :0.0000   Mode  :character   Median :14.45
 Mean   :29.36   Mean   :0.523    Mean   :0.3816                Mean   :32.20
 3rd Qu.:35.00   3rd Qu.:1.0000   3rd Qu.:0.0000                3rd Qu.:31.00
 Max.   :80.00   Max.   :8.0000   Max.   :6.0000                Max.   :512.33

 Embarked
 Length:891
 Class :character
 Mode  :character
```

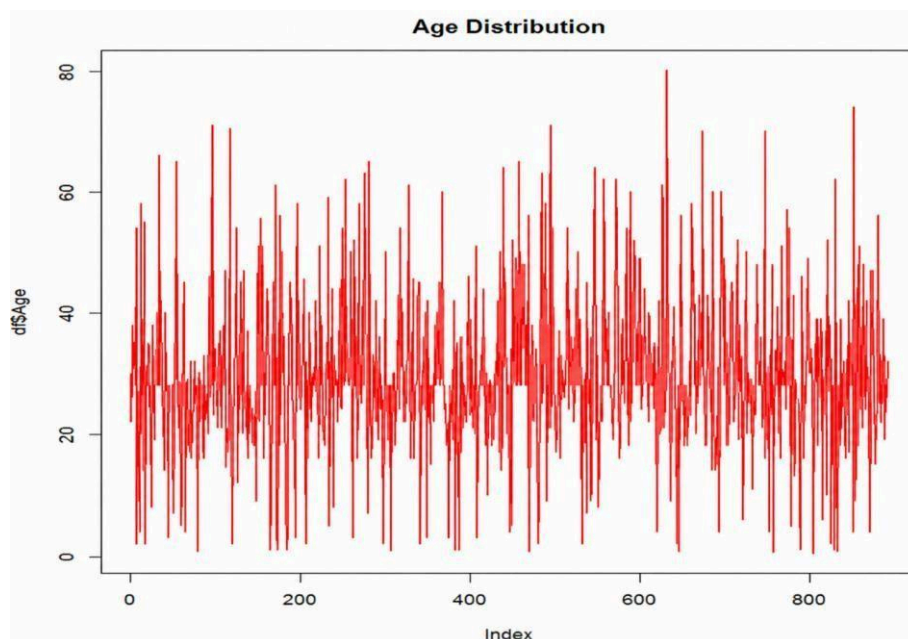
```
> str(df)
'data.frame': 891 obs. of 11 variables:
 $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...
 $ Survived : int 0 1 1 1 0 0 0 0 1 1 ...
 $ Pclass : int 3 1 3 1 3 3 1 3 3 2 ...
 $ Name : chr "Braund, Mr. Owen Harris" "Cummings, Mrs. John Bradley (Florence Briggs Thayer)" "Heikkinen, Miss. Laina" "Futrelle, Mrs. Jacques Heath (Lily May Peel)" ...
 $ Sex : chr "male" "female" "female" "female" ...
 $ Age : num 22 38 26 35 35 28 54 2 27 14 ...
 $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
 $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...
 $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
 $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
 $ Embarked : chr "S" "C" "S" "S" ...
```

3. Write R programs to generate the following visualizations:

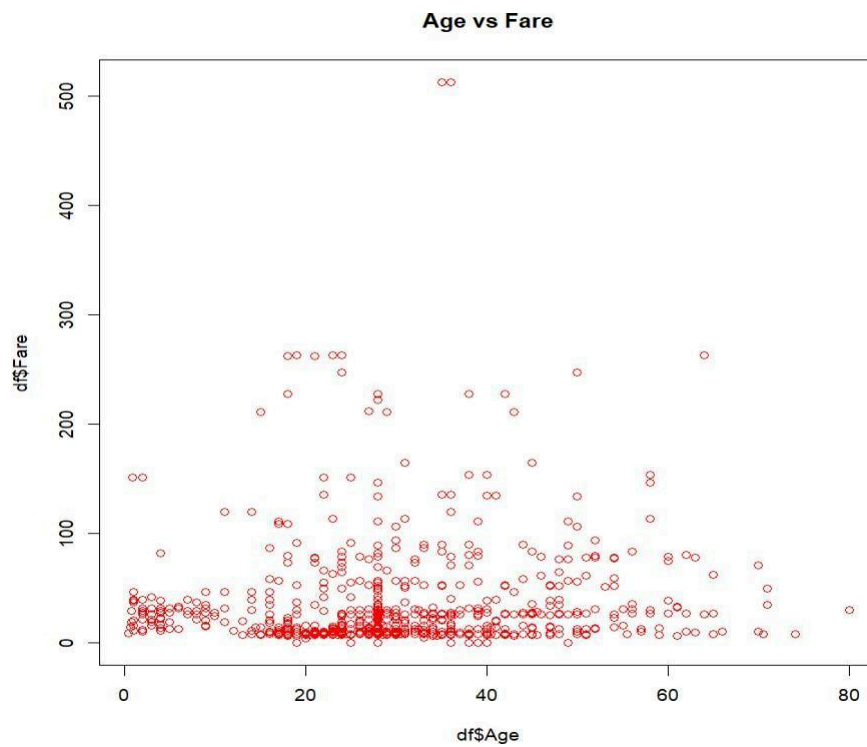
- Bar chart



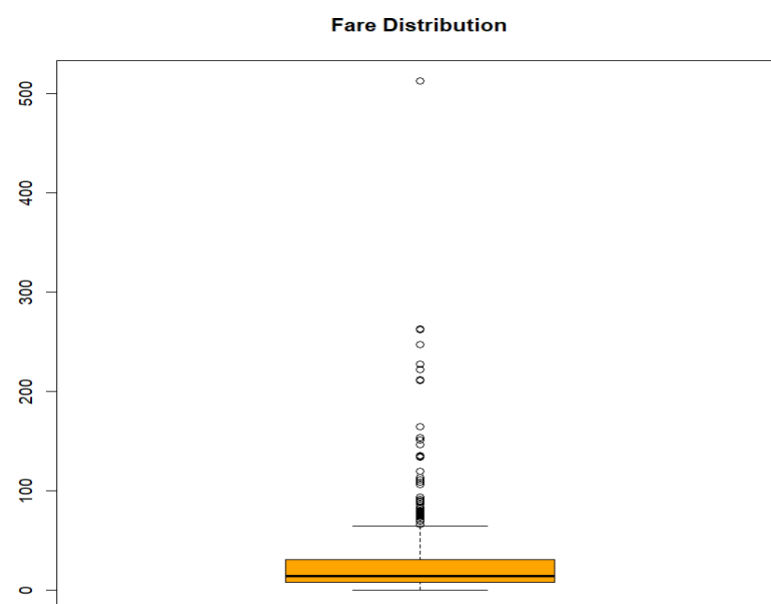
- Line chart



- Scatter plot



- Box plot



4. Perform **Exploratory Data Analysis** on a real dataset using R:

- Summary statistics: `summary(df)`

```
> summary(df)
  PassengerId   Survived  Pclass     Name    Sex
Min.   : 1.0   Min.   :0.0000 Min.   :1.000 Length:891 Length:891
1st Qu.:223.5 1st Qu.:0.0000 1st Qu.:2.000 Class :character Class :character
Median :446.0 Median :0.0000 Median :3.000 Mode  :character Mode  :character
Mean   :446.0 Mean   :0.3838 Mean   :2.309
3rd Qu.:668.5 3rd Qu.:1.0000 3rd Qu.:3.000
Max.   :891.0 Max.   :1.0000 Max.   :3.000

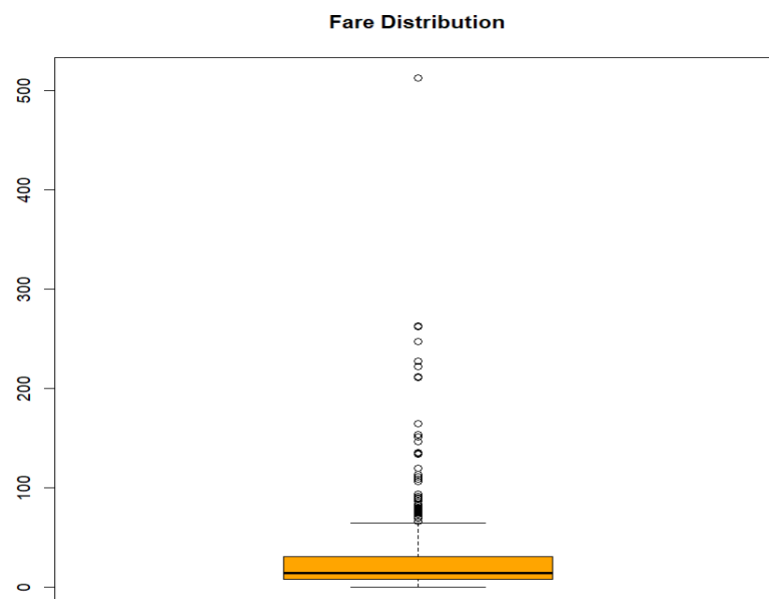
   Age   SibSp   Parch     Ticket     Fare
Min.   : 0.42 Min.   :0.000 Min.   :0.0000 Length:891 Min.   : 0.00
1st Qu.:22.00 1st Qu.:0.000 1st Qu.:0.0000 Class :character 1st Qu.: 7.91
Median :28.00 Median :0.000 Median :0.0000 Mode  :character Median :14.45
Mean   :29.36 Mean   :0.523 Mean   :0.3816          Mean   :32.20
3rd Qu.:35.00 3rd Qu.:1.000 3rd Qu.:0.0000          3rd Qu.:31.00
Max.   :80.00 Max.   :8.000 Max.   :6.0000          Max.   :512.33

  Embarked
Length:891
Class :character
Mode  :character
```

- Correlation analysis: `cor(df[, c("Age", "Fare", "Pclass")])`

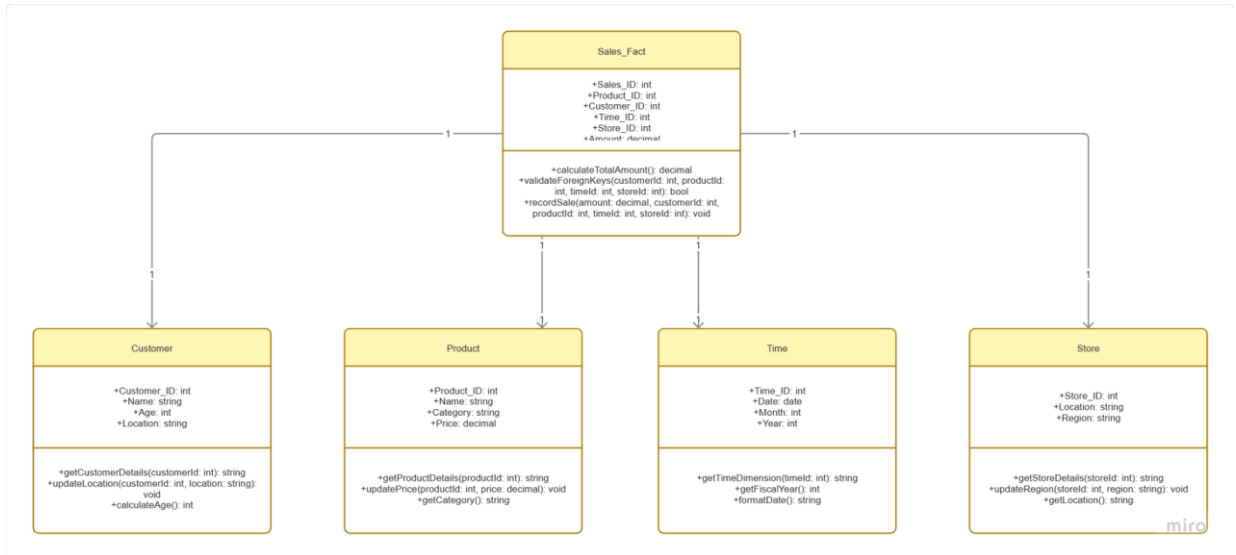
```
> cor(df[, c("Age", "Fare", "Pclass")])
           Age      Fare      Pclass
Age      1.00000000  0.09668842 -0.3398983
Fare      0.09668842  1.00000000 -0.5494996
Pclass   -0.33989833 -0.54949962  1.0000000
```

- Outlier detection: `boxplot(df$Fare, main="Outliers in Fare")`



5. Given a business scenario (Retail / Hospital / University):

- Identify fact and dimension tables
- Design a **dimensional data model**
- Define **KPIs** and explain their business relevance

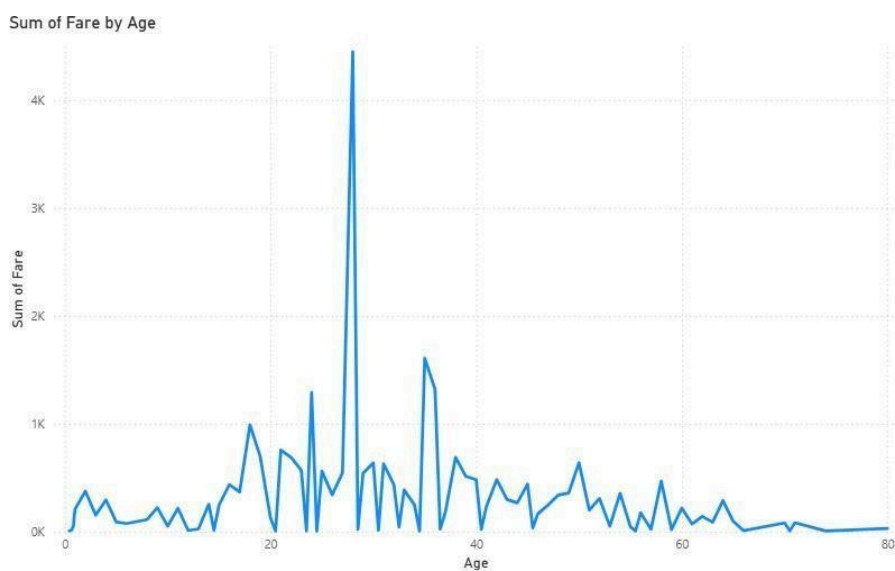
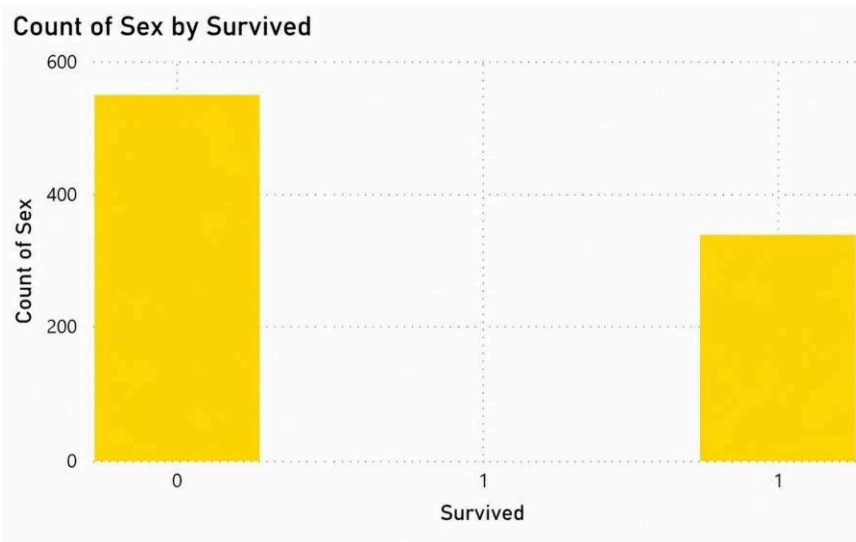


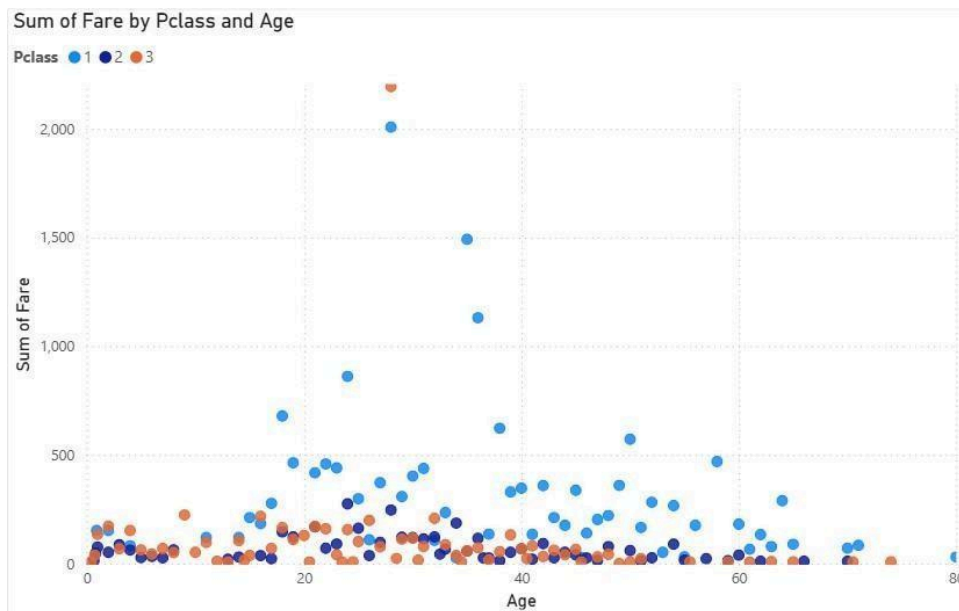
KPIs:

- Total Sales → Business growth
- Revenue per Customer → Customer value
- Top Selling Products → Inventory planning
- Monthly Sales Trend → Forecasting

6. Import a dataset into **Tableau / Power BI** and:

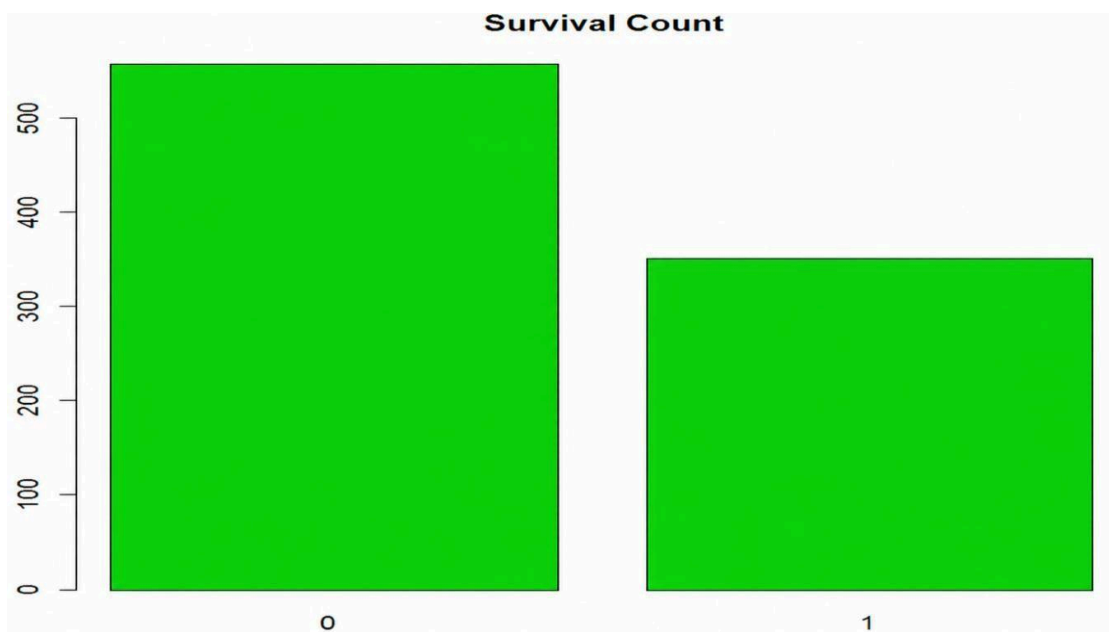
- Identify dimensions and measures
- Create bar, line, and scatter charts





7. Import and visualize data from:

- CSV
- JSON / XML
- SQL or NoSQL database



Age vs Fare